Tutorial - 11

(1) $H_p = g\cos(kx)$.

$E_0' = \langle 0| H_p |0\rangle$.

$= g\langle 0|\cos(kx)|0\rangle$. ~~$= g\cos kx$~~

$$\langle \cos kx\rangle = \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} e^{-\frac{m\omega x^2}{\hbar}} \cdot \cos kx \, dx.$$

$$\left[\, |0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} \,\right].$$

$$= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \mathrm{Re}\int_{-\infty}^{\infty} \cancel{e^{ikx}}\, \cos kx \cdot e^{-\frac{m\omega}{\hbar}x^2} dx.$$

$$\left[\int_{-\infty}^{\infty} \cos kx \, e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}\, e^{-k^2/4\alpha}\right]$$

$$\therefore \cancel{\mathrm{Re}\int_0^{\infty} e^{\left(-\frac{m\omega}{\hbar}x^2 + ikx\right)} dx}$$

$$= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \cdot \sqrt{\frac{\pi\hbar}{m\omega}}\, e^{-\frac{k^2\hbar}{4m\omega}}$$

$$\boxed{\langle \cos kx\rangle = e^{-\hbar k^2/m4m\omega}}$$

$$\therefore E_0' = g e^{-k^2\hbar/4m\omega}. \quad \text{(Ans)}$$

(3) (i) $$H_0 = \begin{pmatrix} 1+\lambda & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & -2\lambda \\ 0 & 0 & -2\lambda & 7 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}}_{H_0} + \lambda \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -2 & 0 \end{pmatrix}}_{H_p}$$

(ii) $E_n' = \langle n| H_p |n\rangle$.

The energy eigenvalues for H_0 are 1, 8, 3 & 7.

$\therefore$ ~~E_0 (Ground~~ E_0 (Ground state) = 1

$E_1 = 3$; $E_2 = 7$; $E_3 = 8$.

$\therefore$ ~~$E_0^1 = \langle 0|H_p|0\rangle$~~. $|0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$; $|1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$; $|2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$;

$|3\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

$\therefore E_0^1 = \langle 0|H_p|0\rangle = \lambda$.

$\therefore E_1^1 = \langle 1|H_p|1\rangle = 0$

$\therefore E_2^1 = \langle 2|H_p|2\rangle = 0$

$\therefore E_3^1 = \langle 3|H_p|3\rangle = 0$

[Ans]

② $E_0^1 = \langle 1s|H'|1s\rangle$.

$= eE \langle 1s|z|1s\rangle$ [$z = r\cos\theta$].

$\langle z\rangle = \int \psi_{1s}^* z \psi_{1s} \, d^3r = A^2 \int_0^\pi \int_0^\infty e^{-2r/a_0} \, r\cos\theta \cdot r^2 \, dr \sin\theta \, d\theta \int_0^{2\pi} d\phi$

[$\psi_{1s} = A e^{-r/a_0}$]

$= A^2 2\pi \int_0^\pi \cos\theta \sin\theta \, d\theta \int_0^\infty e^{-2r/a_0} \cdot r^3 \, dr$.

$\therefore \int_0^\pi \cos\theta \sin\theta \, d\theta = \frac{1}{2} \int_0^\pi \sin 2\theta \, d\theta$

$= -\frac{1}{4} [\cos 2\theta]_0^\pi = 0$.

$\therefore \langle z\rangle = \langle 1s|z|1s\rangle = 0$

$\therefore$ $\boxed{E_0^1 = 0}$ (Ans Proved)